

Market power in an interbank model

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Balance sheet

Following model presented in [3] and [2], We examine a sequential economy that unfolds in discrete time periods, denoted as $t = \{0, 1, 2, \dots, T\}$. At each point in time $t = \{0, 1, 2, \dots, T\}$, the system consists of N banks $i, j \in \Omega = \{1, \dots, N\}$. These financial institutions are linked through the credit relationships captured by the set V_t , whose elements represent ordered pairs of distinct banks. Together, banks (nodes or vertices) and their credit links (edges) constitute the interbank network $G_t = (\Omega, V_t)$. The daily balance sheet composition of each bank is specified as follows:

$$\underbrace{L_t^i + C_t^i + R_t^i}_{\text{assets}} = \underbrace{D_t^i + E_t^i}_{\text{liabilities}} \quad (1)$$

Assets are on the left side and liabilities are on the right side. In particular, L , C , and R represent long-term assets, liquidity, and reserves, while D and E deposits and bank equity i at time t . Reserves are a deposit rate that replicates the central bank regulation that sets the minimum amount that a bank should hold in liquid assets ($R_t^i = \hat{r}D_t^i$), which is used in this model $\hat{r} = 0.02$.

At every time t , deposits are exogenously shocked, and the balance sheet modifies accordingly. Specifically, deposits evolve as

$$D_t^i = D_{t-1}^i[\mu + \omega U(0, 1)] \quad (2)$$

where $U(0, 1)$ is a uniformly distributed noise between 0 and 1 and μ and ω represent the expected number of negative shocks and thus different market conditions. This setup achieves stock-flow equilibrium on average, enabling roughly half of market actors to recoup erosion from negative shocks. While not all strict consistency conditions hold, tests confirm that at $\mu = 0.7$ and $\omega = 0.55$, assets match liabilities per bank, yielding a system-wide balance sheet of zero, as [2] demonstrate.

Banks that suffer an increase in deposits or a small negative shock covered with their own capital become potential creditors to the system, with a supply of liquidity denoted as $s_t^i = \Delta D_t^i + C_t^i$. Those loans are obtained by the banks with a negative change in deposits when $\Delta D_t^i + C_t^i \leq 0$. The demand of liquidity of each borrower is $d_t^i = |\Delta D_t^i + C_t^i|$.

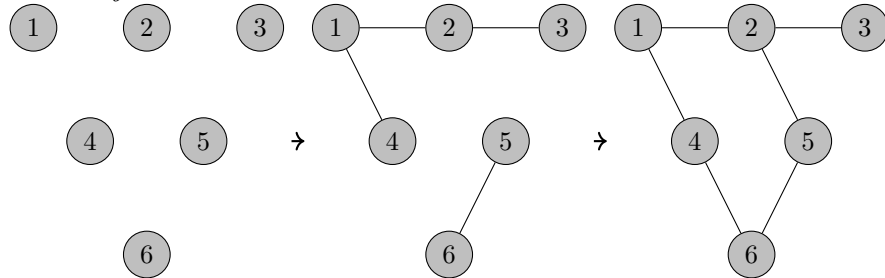
In each t the number of borrowers and lenders are not equalized, as our mechanism 2 is Non-Walrasian. That disequilibrium generates a rationed market increased by the limitation to an only one lender for each borrower, using a mechanism of attachment of lenders and borrowers explained in the next section

As not all borrowers will obtain their full demand for loan d_t^j , we define the real loan obtained for a generic borrower j by its lender i as $l_t^{i,j} = \min(s_t^i, d_t^j)$. The last resort option for the borrower to obtain the amount of money not covered by the loan is to fire sale its long term assets at ρ price, with a reduction in L of $\nabla L_t^j = \frac{d_t^j - s_t^i}{\rho}$.

Mechanism of attachment of lending agreements

At the beginning of each day, agents meet in the interbank market and sign bilateral potential lending agreements representing the directed links $(i, j) \in V_t$. These agreements can be interpreted as credit lines, which are valid during t , and can be used at the request of the borrower j in case of the lender i available liquidity. The set of all potential lending agreements reproduces the potential interbank network and we explicitly limit to one only possible lender for each borrower, so if we represent as a directed graph where nodes are banks and a incoming link in node j from node i represents the potential lending agreement from creditor i for debtor j . The grade of outcoming links is limited to 1, so only one possible potential lender is available.

The preferential lending agreements are determined in each step using a network topology of an Erdős–Rényi random graph $G(N, p_a)$, constructed by connecting initially undirected labeled nodes randomly, in a similar form as presented in [5]. The nodes N are the banks. Each undirected edge is included in the graph with a parameter of probability p_a without any dependency of previous edges, iterating over each combination of possible distinct nodes (i, j) and after a Bernoulli process with probability p_a we add or not the edge connecting nodes i and j :



After the graph is generated, each bank i has L_i connections on the graph, or maybe none, as picture 1 denotes:

The direction of the connections between nodes (banks) is randomly chosen just after the shock to have one lender or no lender (because each borrower can have only one loan). If there are multiple lenders, we randomly choose one of them and set up the loan for this step. When the current time t is

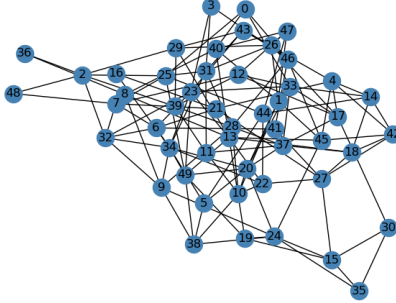


Figure 1: Erdős-Rényi without directions with fifty banks

erdos_renyi p= 0.1 giant=50 comm_not_alone=1 comm=1 gcs=50

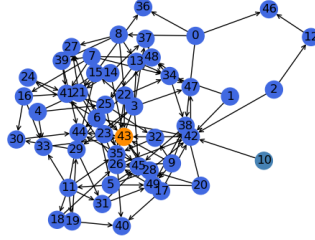


Figure 2: Similar to Erdős-Rényi of 1 but with incoming links representing the banks which could be borrowing from each bank

complete, before starting $t + 1$ a new graph is generated. The different color in 2 of bank 43 is used to denote the one with more incoming links (more active borrowers). As $p_a = 0.1$ there are no isolated banks. Also, the main facts about the network created is added to the graph (such as the Giant Component Size and the number of communities).

Either these links are generated in advance of the shock, they are potential agreements that can or cannot correspond to the real necessities of banks.

Trading strategy

We assume that banks are risk-neutral agents operating in a perfect competition environment to optimize their expected profit. The expected profit of the bank i for a loan provided to j is given by

$$\mathbb{E}[\Pi_t^{i,j}] = \underbrace{p_t^j (r_t^{i,j} c_t^{i,j} - \overbrace{m c_t^{i,j}}^{\text{cost of loan}})}_{\text{normally pays}} - \underbrace{(1 - p_t^j)(m c_t^{i,j})}_{\text{when fails}} \quad (3)$$

where m are the screening cost of the set up the link, defined in this case as $m = 0.015$, and p_t^j is the probability that the borrower will not fail (probability of surviving), $r_t^{i,j}$ the interest rate asked by the lender i to the borrower j , $c_t^{i,j}$ the maximum amount i is willing to lend to j . We assume also that there are no liquidation costs of assets.

Recalling that the borrower fails if its equity becomes negative $E_t^j < 0$, the probability of surviving is given by the closeness between the equity of j and the highest net worth in the system:

$$p_t^j = \frac{E_t^j}{E_t^{max}} \quad p_{b_t}^j = 1 - p_j^t = 1 - \frac{E_t^j}{E_t^{max}} \quad (4)$$

Analogously, we can define the probability of bankruptcy as the opposite, also denoted in 4 as p_b . The probability of surviving the bank is connected to financial fragility. A financial institution leaves the system if its net worth is so low that an adverse shock makes it negative or if she suffers a loss so huge as to deplete all the net worth accumulated in the past. Finally, the maximum amount that the lender i is willing to lend to j , that is the lending capacity is defined as

$$c_t^{i,j} = \begin{cases} (1 - h_t^j) A_t^j > 0, & \text{if } (i, j) \in V_t, \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

where the borrower haircut is $h_t^j \in (0, h_t^{max})$, defined as the j 's leverage, λ_t^j with respect to the maximum one. so, $h_t^j = \frac{\lambda_t^j}{\lambda_t^{max}}$.

The last component we will introduce is the power market ψ . It represents a market power parameter for lenders that captures their bargaining position ($\psi = 0$) or monopolistic power ($\psi = 1$) in our market:

- $\psi = 0$: Competitive level (no lender power). The loan rate is the “fair” or actuarially competitive rate.
- $0 < \psi < 1$: Partial monopolistic power
- $\psi = 1$: Monopoly. Lenders use market power (e.g., limited competition, switching costs, informational advantages) to inflate the rate above this competitive benchmark, proportionally to a double in the competitive level $(1+\psi)$.

This parameter ψ captures information asymmetries (lenders' advantage over borrowers in a non-competitive market) or the opposite desired situation with no lenders charging high rates and borrowers easily switching lenders because there are many connections.

With all these components, and doing equal to zero the expected profit in 3:

$$\begin{aligned}
\mathbb{E}[\Pi_t^{i,j}] &= p_t^j (r_t^{i,j} c_t^{i,j} - m c_t^{i,j}) - (1 - p_t^j)(m c_t^{i,j}) &= 0 \\
p_t^j r_t^{i,j} c_t^{i,j} &= (1 - p_t^j)(m c_t^{i,j}) + p_t^j m c_t^{i,j} \\
r_t^{i,j} &= \frac{m c_t^{i,j} - p_t^j m c_t^{i,j} + p_t^j m c_t^{i,j}}{p_t^j c_t^{i,j}} \\
r_t^{i,j} &= \frac{m c_t^{i,j} - \cancel{p_t^j m c_t^{i,j}} + \cancel{p_t^j m c_t^{i,j}}}{p_t^j c_t^{i,j}} \\
r_t^{i,j} &= \frac{m \cancel{c_t^{i,j}}}{p_t^j \cancel{c_t^{i,j}}} \\
r_t^{i,j} &= \frac{m}{p_t^j}
\end{aligned}$$

Now we can introduce our power market, having the interest rate formula as

$$\begin{aligned}
r_t^{i,j} &= \underbrace{\frac{m}{p_t^j}}_{\text{baseline}} \times \underbrace{\frac{1}{(1 - \psi)}}_{\text{lender markup}} \\
r_t^{i,j} &= \frac{m}{p_t^j (1 - \psi)}
\end{aligned}$$

Consistent with asymmetric information and costly state verification [1], interest rates are determined by the size of the lender (measured by assets) and the leverage of the borrower. The budget identity underpins this pricing mechanism.

Execute the model with a similar to a Robust Scaler algorithm normalization.

The time series of the interest rate obtained is affected with high kurtosis. A normalization is implemented using the Robust Scaler formula (subtracting the median and divides by the Interquartile Range, a quantile-based min-max normalization applied per time step) but with some adaptations. Here is the formulation:

1. Let determine first the set of interest rates at time t :

$$\mathcal{I}_t = \{r_i \mid r_i \in \mathbb{R}, i = 1, \dots, N_t\}$$

2. We compute the low and high quantiles of this set ($p_{\text{low}} = 5\%$ and $p_{\text{high}} = 95\%$ by default):

$$q_{\text{low}} = Q_{p_{\text{low}}}(\mathcal{I}_t), \quad q_{\text{high}} = Q_{p_{\text{high}}}(\mathcal{I}_t).$$

3. We clip the outliers:

$$\tilde{r}_i = \max(q_{\text{low}}, \min(q_{\text{high}}, r_i))$$

4. And min-max normalize into $[0, R]$ (with $R = 1.0$ by default):

$$r_i^{(\text{robust})} = \begin{cases} R \frac{\tilde{r}_i - q_{\text{low}}}{q_{\text{high}} - q_{\text{low}}}, & q_{\text{high}} > q_{\text{low}} \\ 0, & q_{\text{high}} = q_{\text{low}} \end{cases}$$

Equity evolution

The next day, the repayment round takes place. Financial institutions encounter a new deposit movement that increases or decreases their liquidity once again 2. A new change in deposits face the banks to be able to pay the previous loan with the new income (positive shock after an initial negative), have a new opportunity to borrow the extra earns or directly bankruptcy if after the fire-sale process long term assets can not cover the debts in the worst scenario (negative shock after another previous negative one):

- If the borrower has a positive shock the liquidity could repay both principal and the interest
- If the positive shock has not enough impact in deposits or if it is a negative shock, the borrower must sell its long-term assets to pay the principal, the interest and also the possible new negative variation in deposits, or fail if the long-term assets are not enough to pay the obligations, generating a bad debt in each lender bank i due to the borrower j noticed as $B_t^{i,j}$.
- If the borrower j pays to lender i the loan correctly loan $l^{i,j}$, it generates earns at the interest rate $r^{i,j}$, so the equity of each bank evolves as (θ is the subset of failed banks and $\hat{L}$ the fire sales):

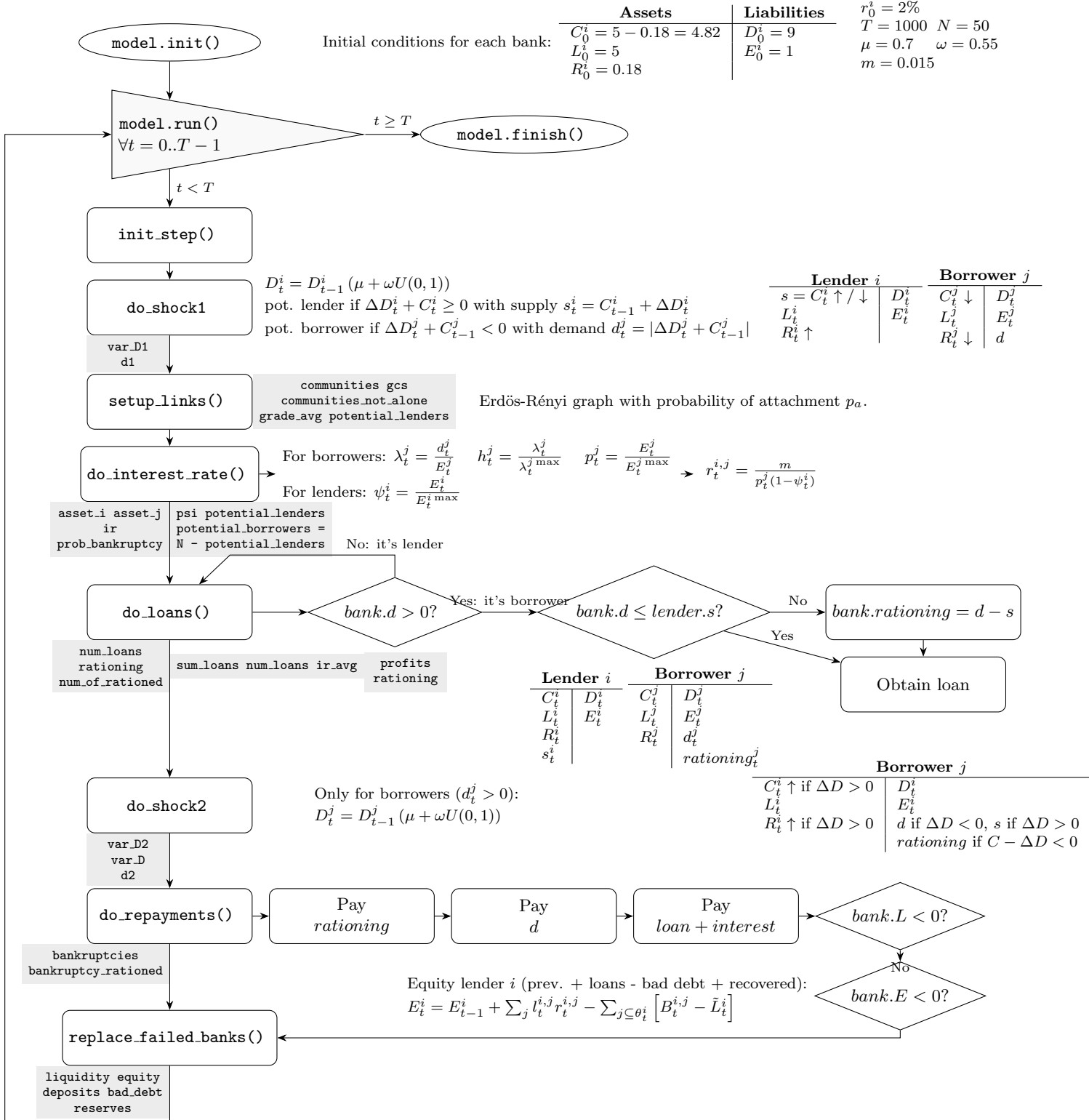
Banks that are failing affect creditors, as the law of motion of their equity is:

$$E_t^i = E_{t-1}^i + \underbrace{\sum_j l_{t-1}^{i,j} r_{t-1}^{i,j}}_{\text{repayment}} - \underbrace{\sum_{j \subseteq \theta_t^i} B_t^{i,j} - \tilde{L}_t^i}_{\text{bad debt}} \quad (6)$$

where we add to the equity of each bank the repayments of returned loans, using the specific interest rate $r^{i,j}$ for each granted loan $l^{i,j}$. And we subtract from the equity the bad debt generated by failed banks $j \subseteq \theta_t^i$ except the amount recovered by the fire sales. Banks that cannot repay the principal and interest in full or cover their necessities also doing fire sales, exit the market ($E_t^i < 0$). Banks exiting in t are replaced in $t + 1$ by the same number of entrants as outgoings, Entrants' size is small, as the initial values are used for these incumbents.

Table 1: Constants used in the simulations

Constant	Value	Description
μ	0.7	Shock parameters for perfect balance
ω	0.55	
m	0.015	
L_{i_0}	5	Long term assets initial value
C_{i_0}	5	Initial Capital.
		After reserves estimation it will be 4.82 (as $R_{i_0} = 0.18$)
D_{i0}	9	Initial Deposits value for each firm
E_{i0}	1	Initial Equity initial
r_i0	0.02	Initial rate



How network density shapes systemic risk

The literature has evolved from traditional bank runs, understood by Diamond and Dybvig as self-fulfilling panics, toward more complex mechanisms:

- Asset price contagion: propagation through the devaluation of collateral in the market.
- Interlocking exposures: the contemporary approach in which the network of direct obligations among financial institutions acts as the main vector of systemic insolvency.

While the traditional view suggested that full connectivity (100%) minimizes risk through risk sharing, more recent research [4] shows a non-monotonic relationship:

- Low connectivity: increasing links stabilizes the system by diluting idiosyncratic shocks.
- High connectivity: once a critical threshold is exceeded, interdependence facilitates the propagation of disturbances, transforming the mutual insurance mechanism into a channel for widespread contagion.

The model presented in this document fits in a high connected network (Erdős–Rényi graph) in which we can study the propagation of failures and resilience.

If we run a simulation through the network connectivity parameter p :

$$p \in \{0.0075, 0.01, 0.05, 0.07, 0.08, 0.09, 0.35, 0.5, 0.9\}$$

For each p_k in the list, a Monte Carlo set of executions $\mathcal{MC} = [1..10]$, running the full T -step simulations and collecting the $(T \times V)$ output (where V is the number of output variables). For each output variable v , compute the time-and-run average and the standard deviation across the runs:

$$\bar{v}_k = \frac{1}{10T} \sum_{m=1}^{10} \sum_{t=0}^{T-1} v_{m,t}, \quad \sigma_{v,k} = \text{std}(\{v_{m,t}\}_{m=1,\dots,10; t=0,\dots,T-1}).$$

All the plots from now are displaying $\bar{v}_k$ with standard deviation p_k . The GCS (Giant Component Size) of the network of banks displays the proportion of the banks that belong to the single dominant group. In an Erdős–Rényi model the transition around $p \approx 0.08$ corresponds to the percolation threshold of the graph, given by the condition on the average degree:

$$\langle k \rangle = p(N-1) \approx 1,$$

In Figure 3 we can observe the change with different values of $p = [0..1]$.

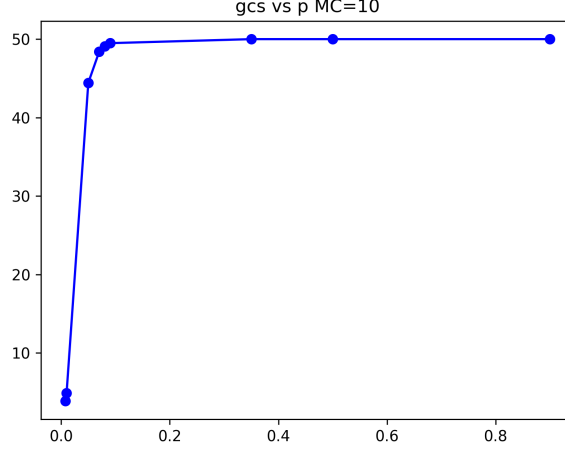


Figure 3: GCS evolution of the model

Interest rate

There is an opposite behavior below the phase transition between results with and without normalization, because it is applied *per time step*, not globally over the simulation (Figure 4).

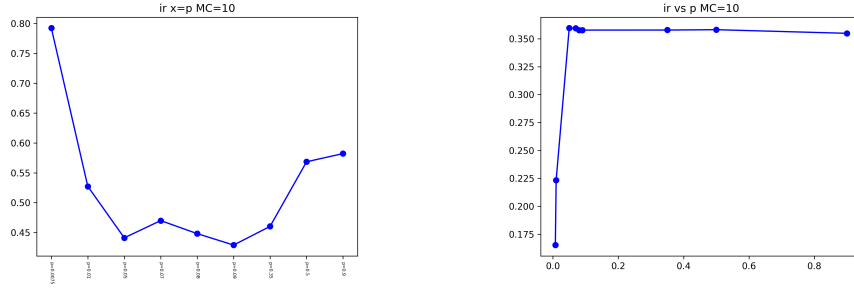


Figure 4: Difference in interest rate averages with and without normalization

Without normalization, for small p , few banks get loans, and those that do are highly distressed (low equity $\Rightarrow$ low probability of bankruptcy $\Rightarrow$ very high interest rate). The mean high interest rates. For large p , more links and competition lead to borrowers with better equity, so the interest decreases.

With normalization, for small p , there are few borrowers and one has extreme interest rates, so most values map near 0 and the mean is low (e.g. ~ 0.2 if range = 1). For large p , ir values are more evenly distributed, so normalization spreads them across $[0, \text{range}]$ and the mean approaches range/2. As a result,

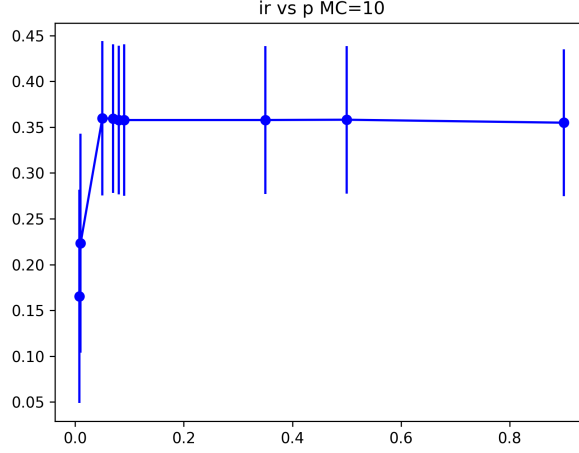


Figure 5: Interest rate and p

the mean increases with p (from ~ 0.2 to ~ 0.5), opposite to the non-normalized case.

Considering so only the results in our normalized model, the consequences are that:

- For low p : the network is sparse, so many borrowers are rationed; this produces *concentrated failures*, leading to high numbers of bankruptcies and a high mean interest rate (because the few loans that occur are distress loans).
- For high p (roughly $p \gtrsim 0.08$): the graph becomes nearly complete (the giant component size approaches N), so borrowers easily find lenders; failures drop, the mean interest rate falls due to competitive pricing, and bad debt shrinks.
- The transition around $p \approx 0.08$ is what we will refer as **phase transition**. The interest rate is positive correlated with ψ (market power), p_b (probability of bankruptcy), bankruptcies and not significant but also with bad debt.

Other considerations of the model

- Bad debt is small in this model because failed banks are immediately replaced. A failed bank loses its (L) assets to pay the lender, then resets to initial conditions next step.
- Bankruptcies are observed with same phase transition as interest rate:

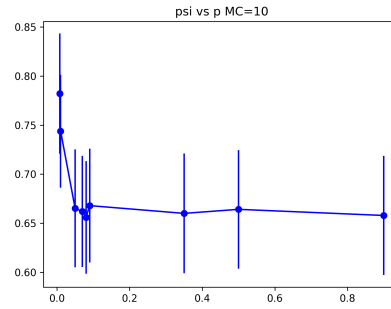


Figure 6: Enter Caption

- Market power, denoted as ψ , is correlated with our interest rate and decreasing with the increment of p : as more links are set up between banks, less power market is possible as competitors start to offer more loans.

References

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